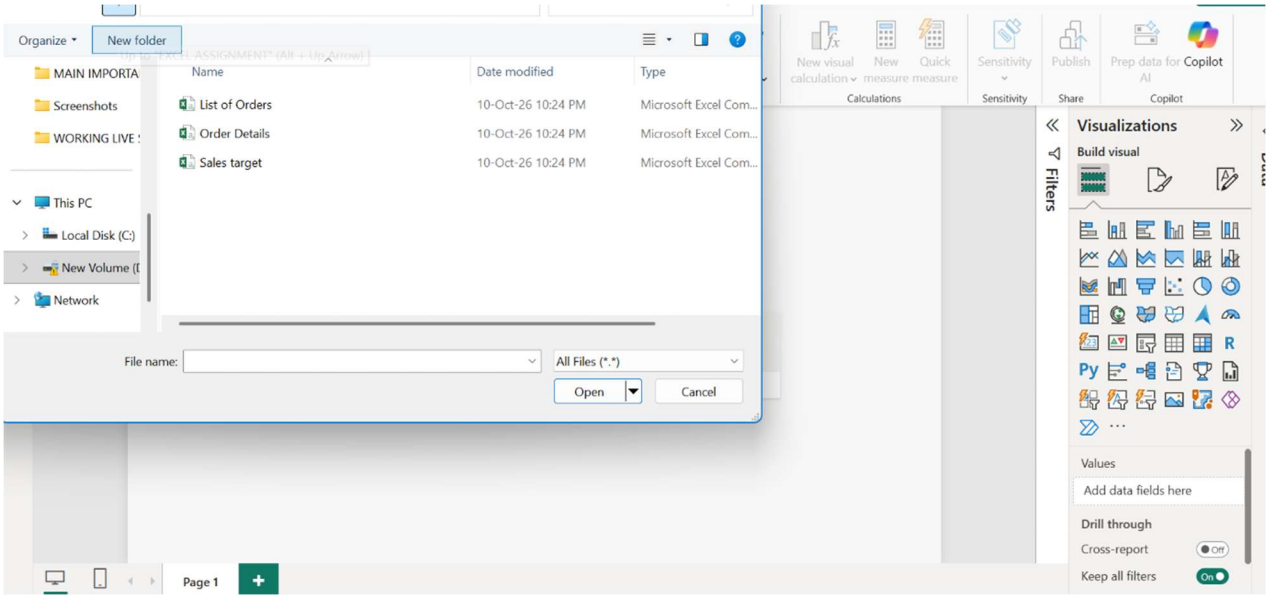
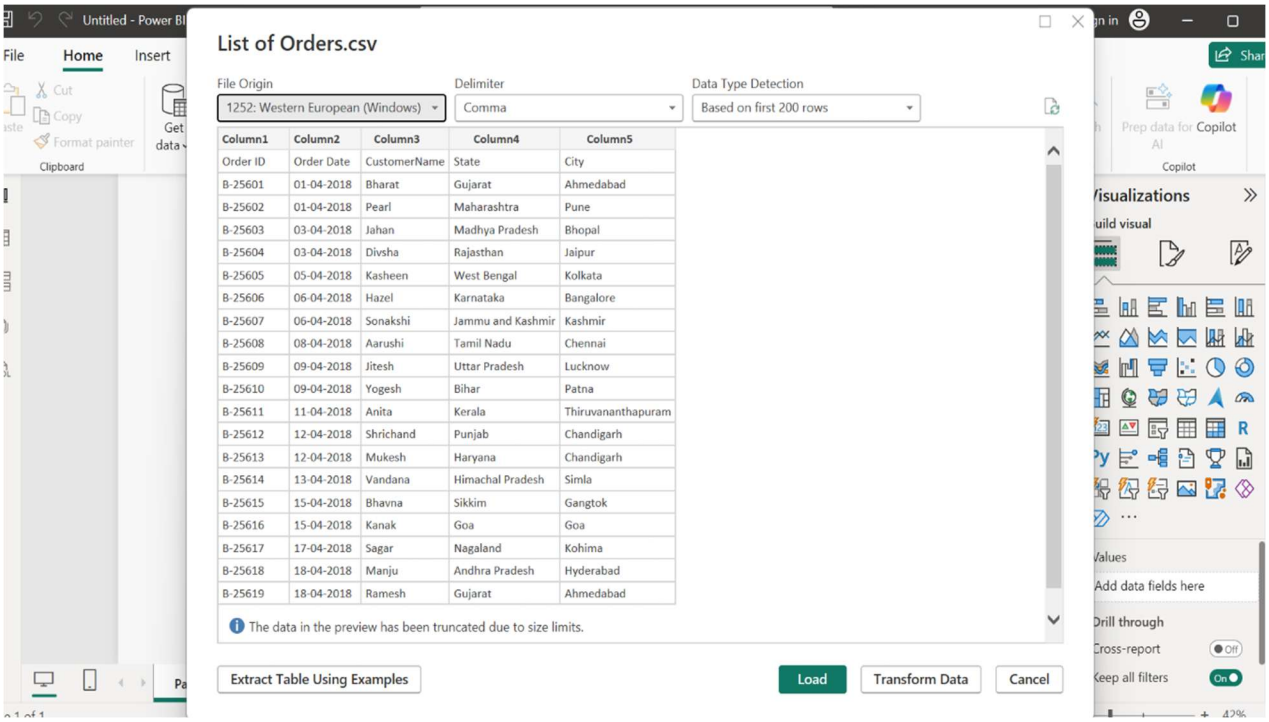
Data Import & Transformation

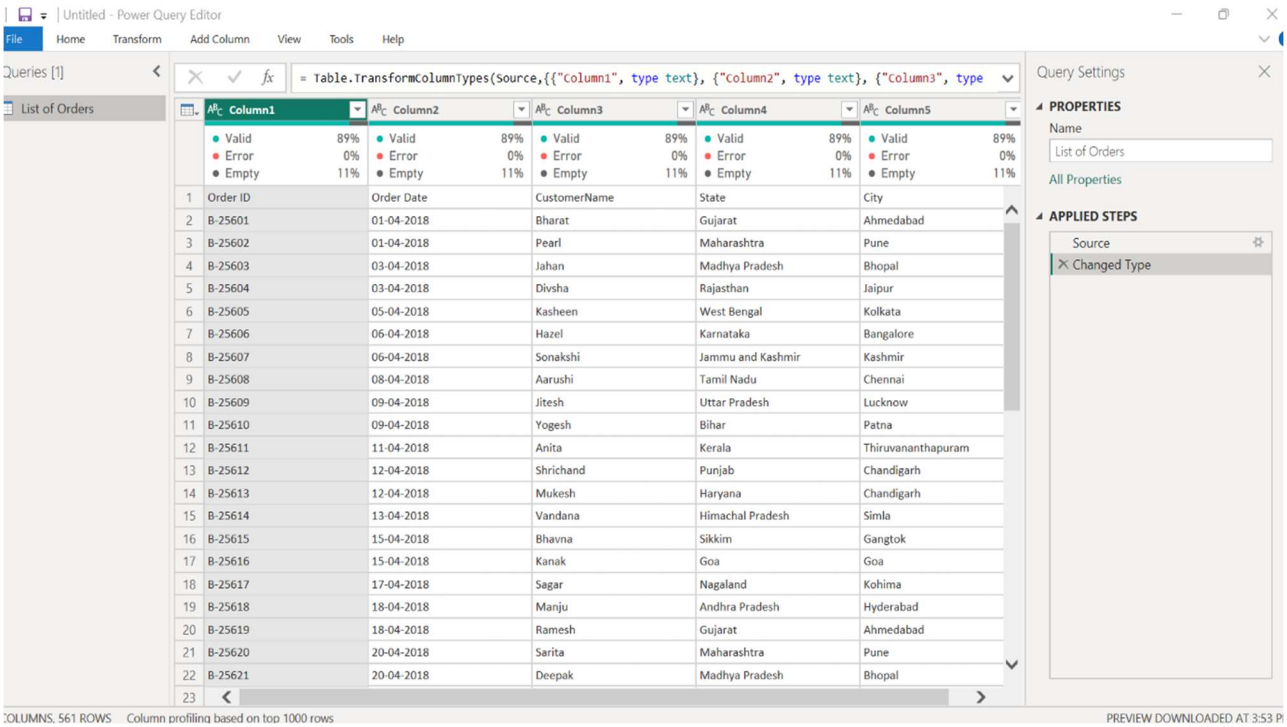
Import List of Orders.csv into Power BI:



List of Orders.csv Transform Data:



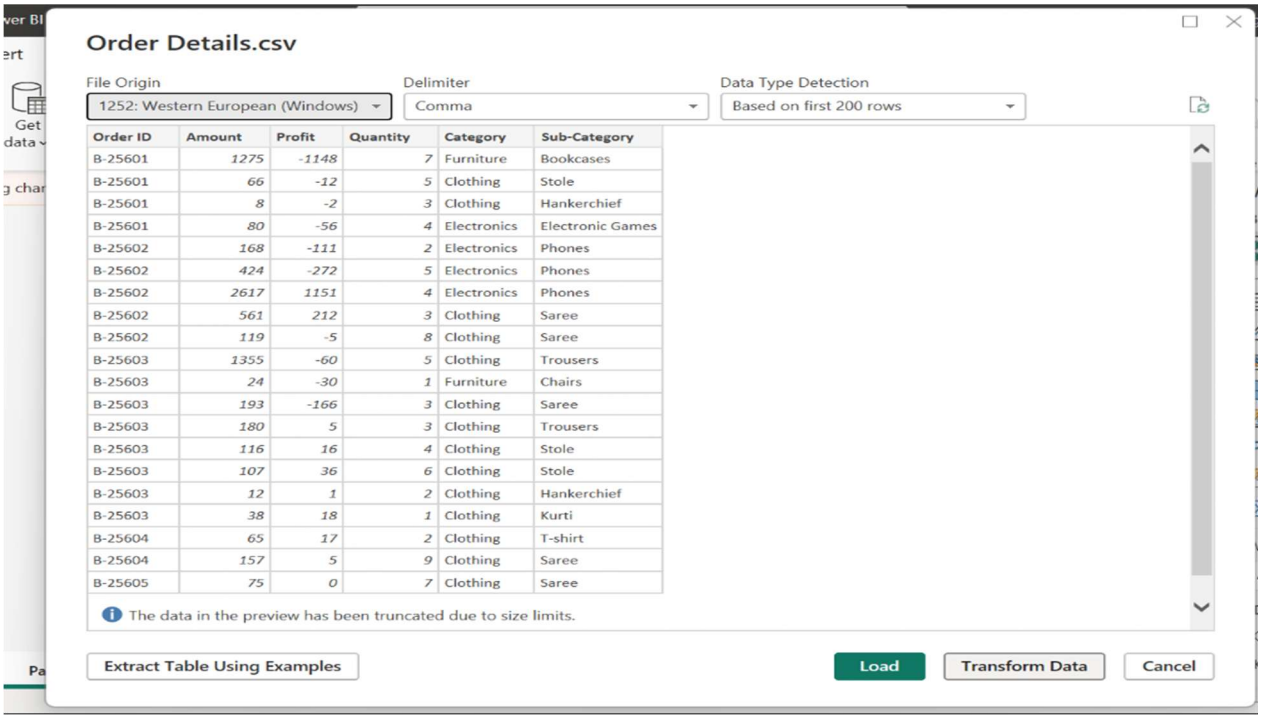
List of orders in the Power Query Editor using the Transform Data:



	Column1	Column2	Column3	Column4	Column5
	Valid 89% Error 0% Empty 11%	Valid 89% Error 0% Empty 11%	Valid 89% Error 0% Empty 11%	Valid 89% Error 0% Empty 11%	Valid 89% Error 0% Empty 11%
1	Order ID	Order Date	CustomerName	State	City
2	B-25601	01-04-2018	Bharat	Gujarat	Ahmedabad
3	B-25602	01-04-2018	Pearl	Maharashtra	Pune
4	B-25603	03-04-2018	Jahan	Madhya Pradesh	Bhopal
5	B-25604	03-04-2018	Divsha	Rajasthan	Jaipur
6	B-25605	05-04-2018	Kasheen	West Bengal	Kolkata
7	B-25606	06-04-2018	Hazel	Karnataka	Bangalore
8	B-25607	06-04-2018	Sonakshi	Jammu and Kashmir	Kashmir
9	B-25608	08-04-2018	Aarushi	Tamil Nadu	Chennai
10	B-25609	09-04-2018	Jitesh	Uttar Pradesh	Lucknow
11	B-25610	09-04-2018	Yogesh	Bihar	Patna
12	B-25611	11-04-2018	Anita	Kerala	Thiruvananthapuram
13	B-25612	12-04-2018	Shrichand	Punjab	Chandigarh
14	B-25613	12-04-2018	Mukesh	Haryana	Chandigarh
15	B-25614	13-04-2018	Vandana	Himachal Pradesh	Simla
16	B-25615	15-04-2018	Bhavna	Sikkim	Gangtok
17	B-25616	15-04-2018	Kanak	Goa	Goa
18	B-25617	17-04-2018	Sagar	Nagaland	Kohima
19	B-25618	18-04-2018	Manju	Andhra Pradesh	Hyderabad
20	B-25619	18-04-2018	Ramesh	Gujarat	Ahmedabad
21	B-25620	20-04-2018	Sarita	Maharashtra	Pune
22	B-25621	20-04-2018	Deepak	Madhya Pradesh	Bhopal
23					

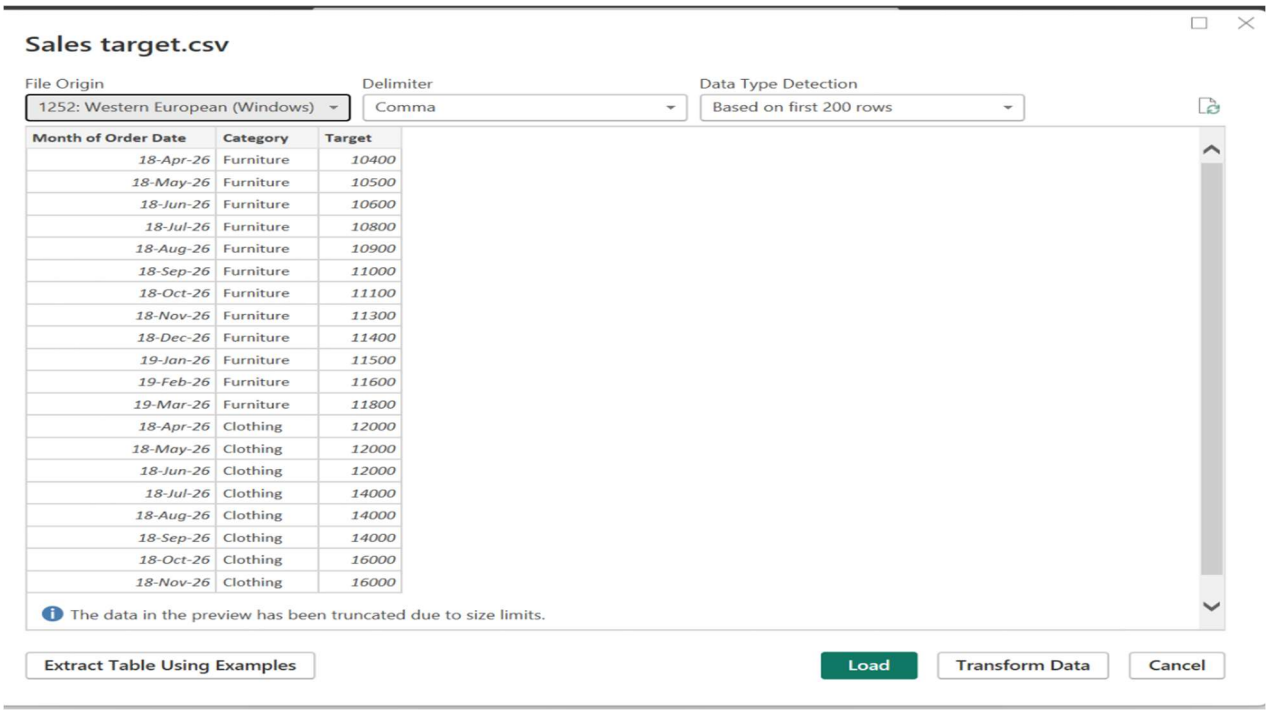
Import Order Details.csv and Sales Target.csv into Power Query Editor:

<u>Order Details.csv import and Transform:</u>

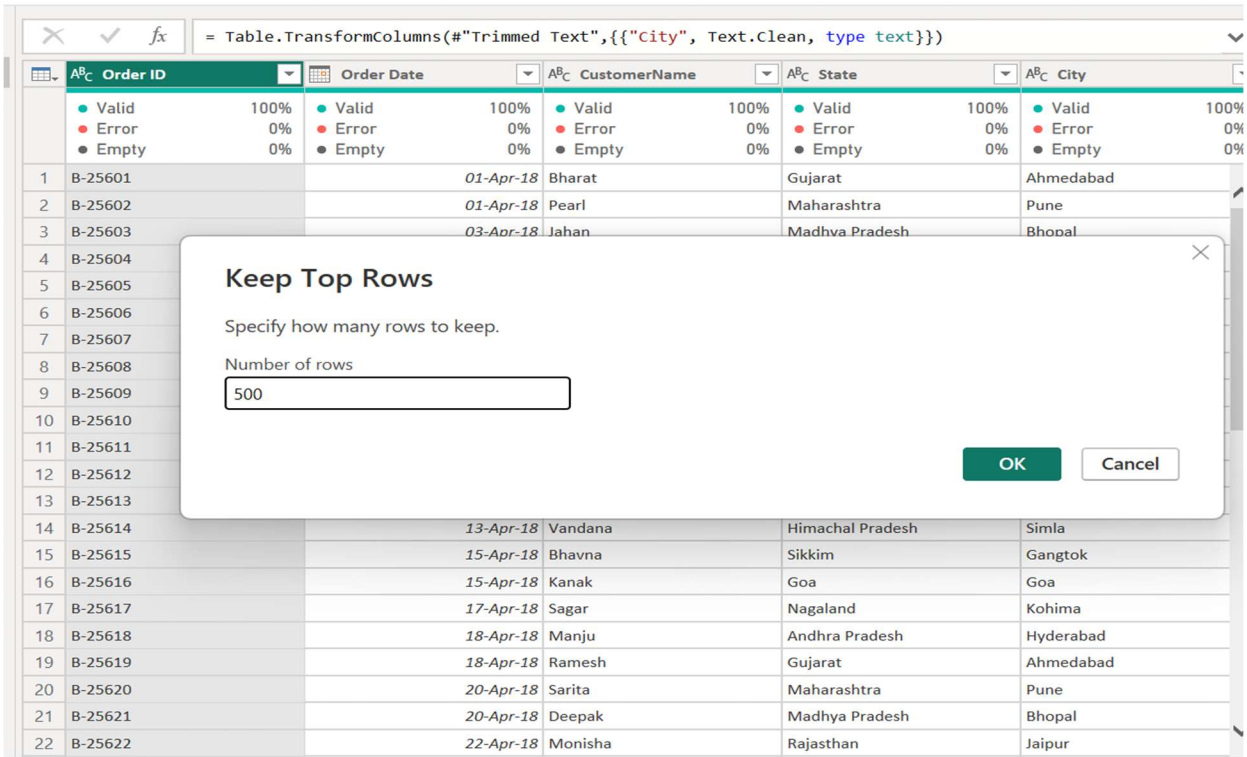


Order ID	Amount	Profit	Quantity	Category	Sub-Category
B-25601	1275	-1148	7	Furniture	Bookcases
B-25601	66	-12	5	Clothing	Stole
B-25601	8	-2	3	Clothing	Hankerchief
B-25601	80	-56	4	Electronics	Electronic Games
B-25602	168	-111	2	Electronics	Phones
B-25602	424	-272	5	Electronics	Phones
B-25602	2617	1151	4	Electronics	Phones
B-25602	561	212	3	Clothing	Saree
B-25602	119	-5	8	Clothing	Saree
B-25603	1355	-60	5	Clothing	Trousers
B-25603	24	-30	1	Furniture	Chairs
B-25603	193	-166	3	Clothing	Saree
B-25603	180	5	3	Clothing	Trousers
B-25603	116	16	4	Clothing	Stole
B-25603	107	36	6	Clothing	Stole
B-25603	12	1	2	Clothing	Hankerchief
B-25603	38	18	1	Clothing	Kurti
B-25604	65	17	2	Clothing	T-shirt
B-25604	157	5	9	Clothing	Saree
B-25605	75	0	7	Clothing	Saree

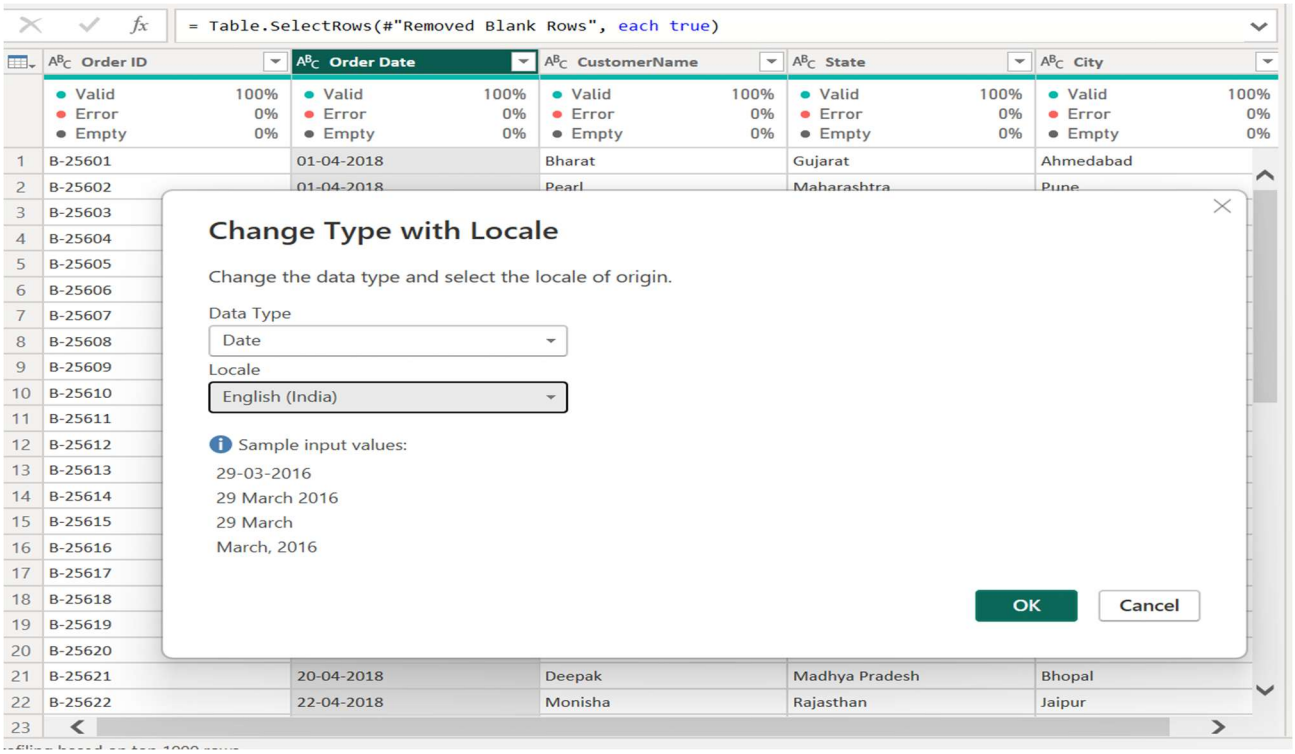
Sales Target.csv import and Transform:



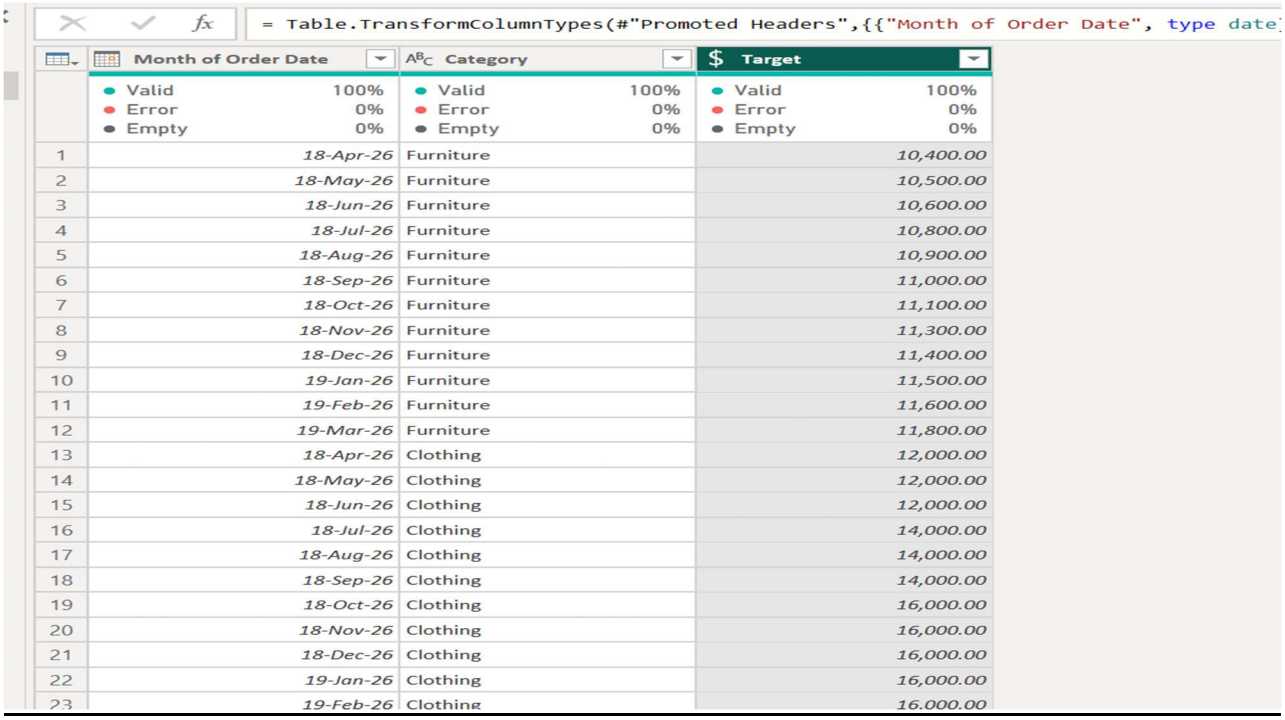
List of Orders table to the first 500 rows:



Set the Order Date column to the Date data type:

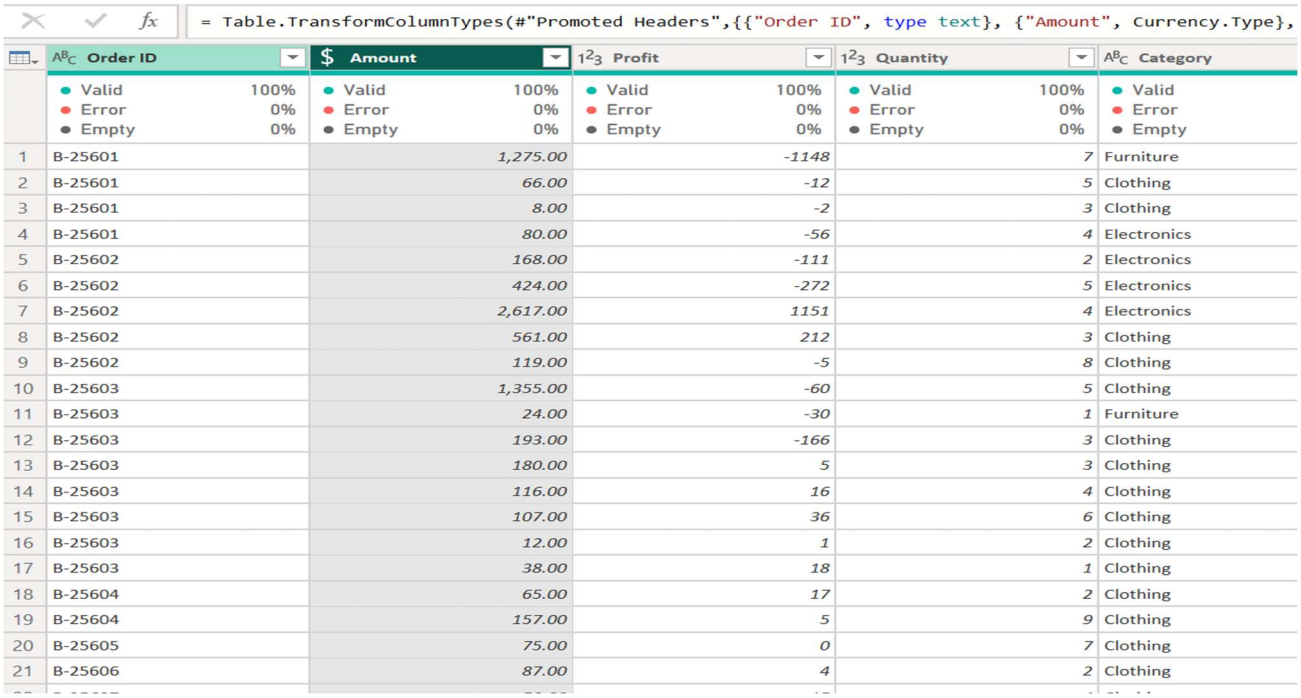


Change the Amount and Target columns to fixed decimal numbers:



	Month of Order Date	Category	Target
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	18-Apr-26	Furniture	10,400.00
2	18-May-26	Furniture	10,500.00
3	18-Jun-26	Furniture	10,600.00
4	18-Jul-26	Furniture	10,800.00
5	18-Aug-26	Furniture	10,900.00
6	18-Sep-26	Furniture	11,000.00
7	18-Oct-26	Furniture	11,100.00
8	18-Nov-26	Furniture	11,300.00
9	18-Dec-26	Furniture	11,400.00
10	19-Jan-26	Furniture	11,500.00
11	19-Feb-26	Furniture	11,600.00
12	19-Mar-26	Furniture	11,800.00
13	18-Apr-26	Clothing	12,000.00
14	18-May-26	Clothing	12,000.00
15	18-Jun-26	Clothing	12,000.00
16	18-Jul-26	Clothing	14,000.00
17	18-Aug-26	Clothing	14,000.00
18	18-Sep-26	Clothing	14,000.00
19	18-Oct-26	Clothing	16,000.00
20	18-Nov-26	Clothing	16,000.00
21	18-Dec-26	Clothing	16,000.00
22	19-Jan-26	Clothing	16,000.00
23	19-Feb-26	Clothing	16,000.00

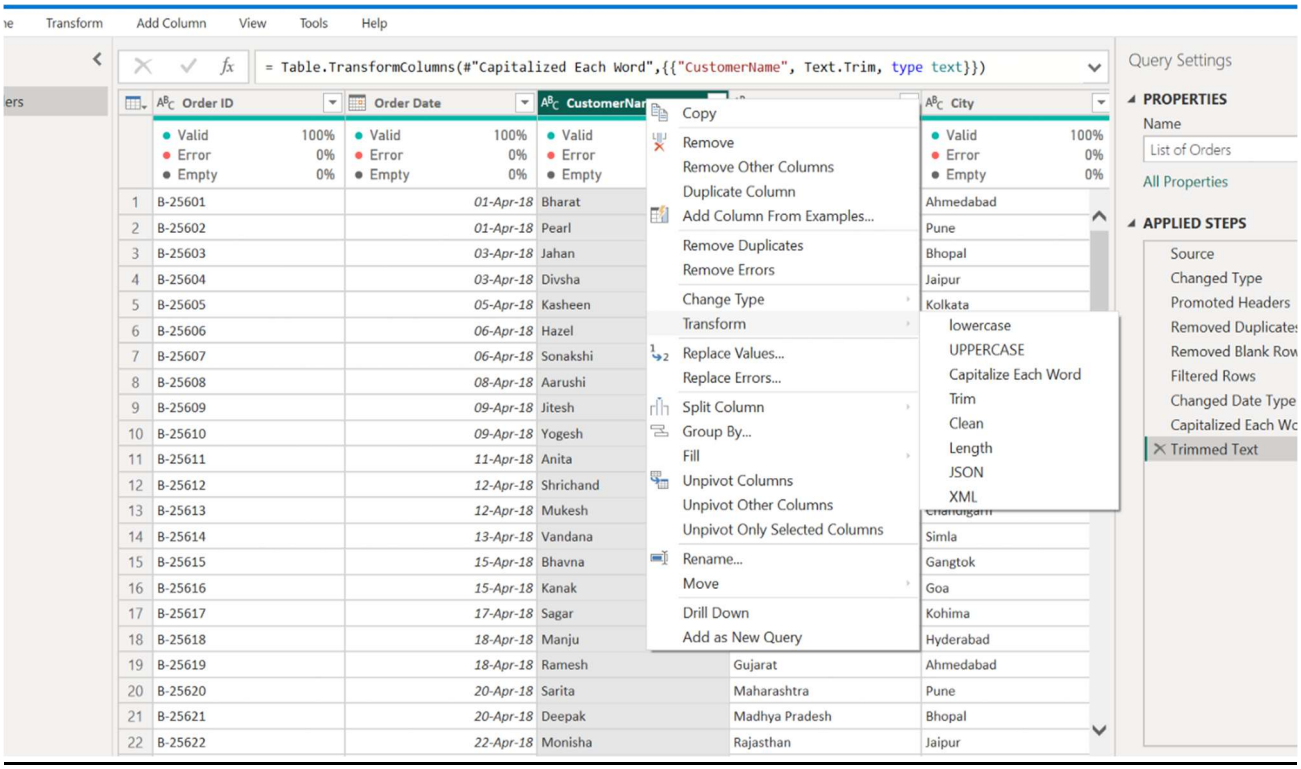
Amount column to fixed decimal Numbers:



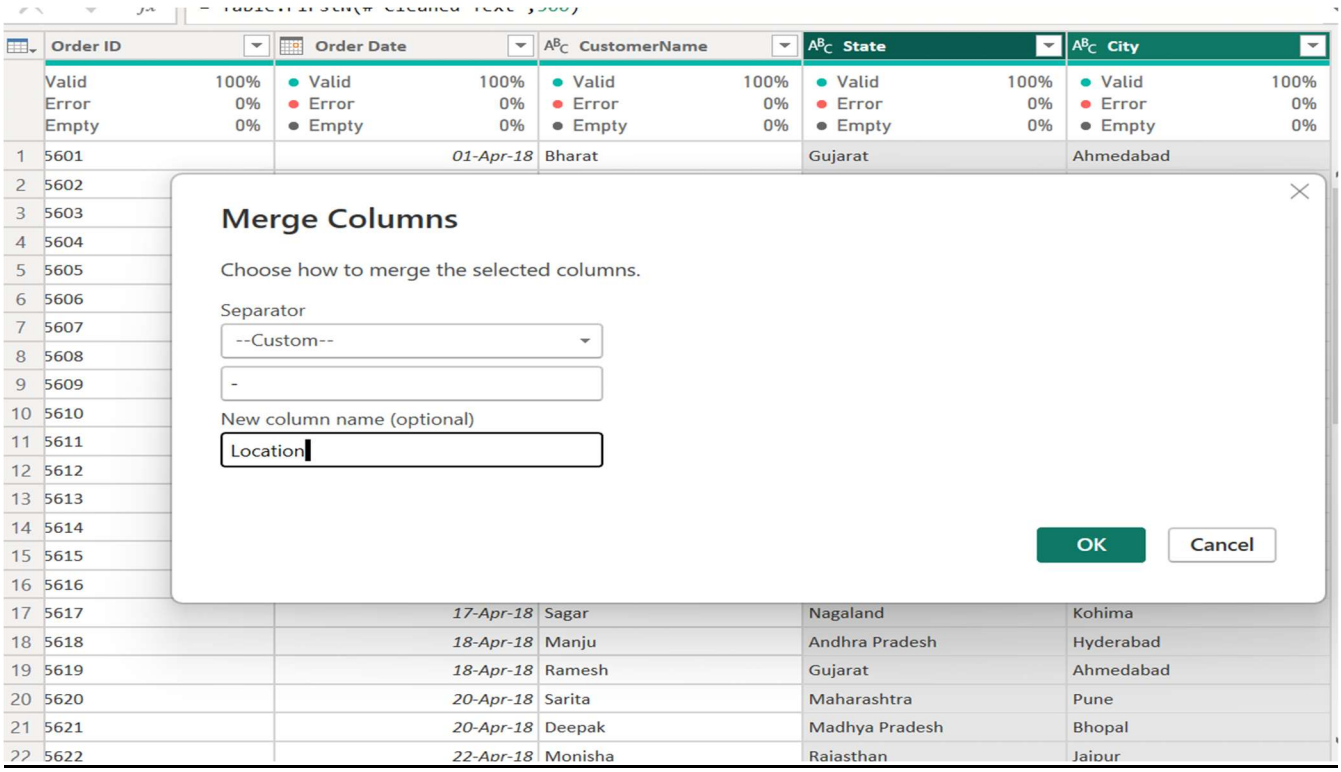
```
= Table.TransformColumnTypes(#"Promoted Headers",{{"Order ID", type text}, {"Amount", Currency.Type},
```

	Order ID	Amount	Profit	Quantity	Category
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid Error Empty
1	B-25601	1,275.00	-1148	7	Furniture
2	B-25601	66.00	-12	5	Clothing
3	B-25601	8.00	-2	3	Clothing
4	B-25601	80.00	-56	4	Electronics
5	B-25602	168.00	-111	2	Electronics
6	B-25602	424.00	-272	5	Electronics
7	B-25602	2,617.00	1151	4	Electronics
8	B-25602	561.00	212	3	Clothing
9	B-25602	119.00	-5	8	Clothing
10	B-25603	1,355.00	-60	5	Clothing
11	B-25603	24.00	-30	1	Furniture
12	B-25603	193.00	-166	3	Clothing
13	B-25603	180.00	5	3	Clothing
14	B-25603	116.00	16	4	Clothing
15	B-25603	107.00	36	6	Clothing
16	B-25603	12.00	1	2	Clothing
17	B-25603	38.00	18	1	Clothing
18	B-25604	65.00	17	2	Clothing
19	B-25604	157.00	5	9	Clothing
20	B-25605	75.00	0	7	Clothing
21	B-25606	87.00	4	2	Clothing

Format the Customer Name column to Proper case:



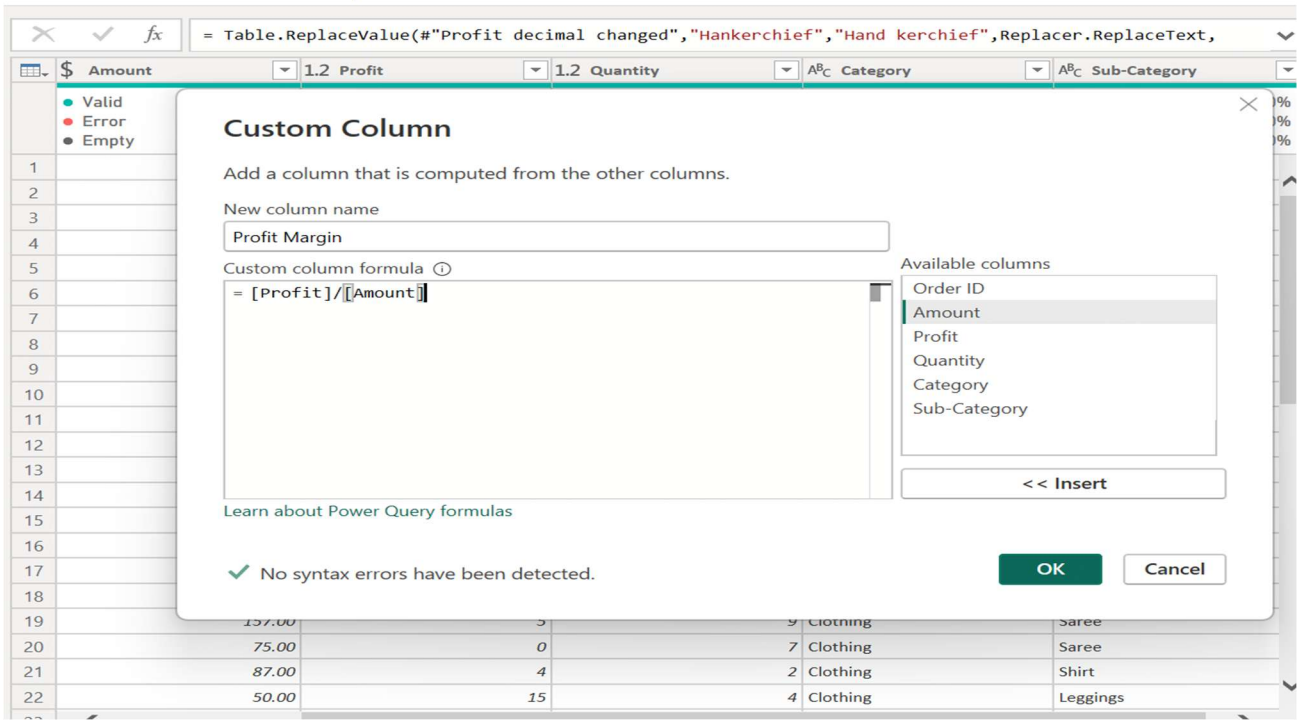
Merge the City and State columns to create a new column named Location:

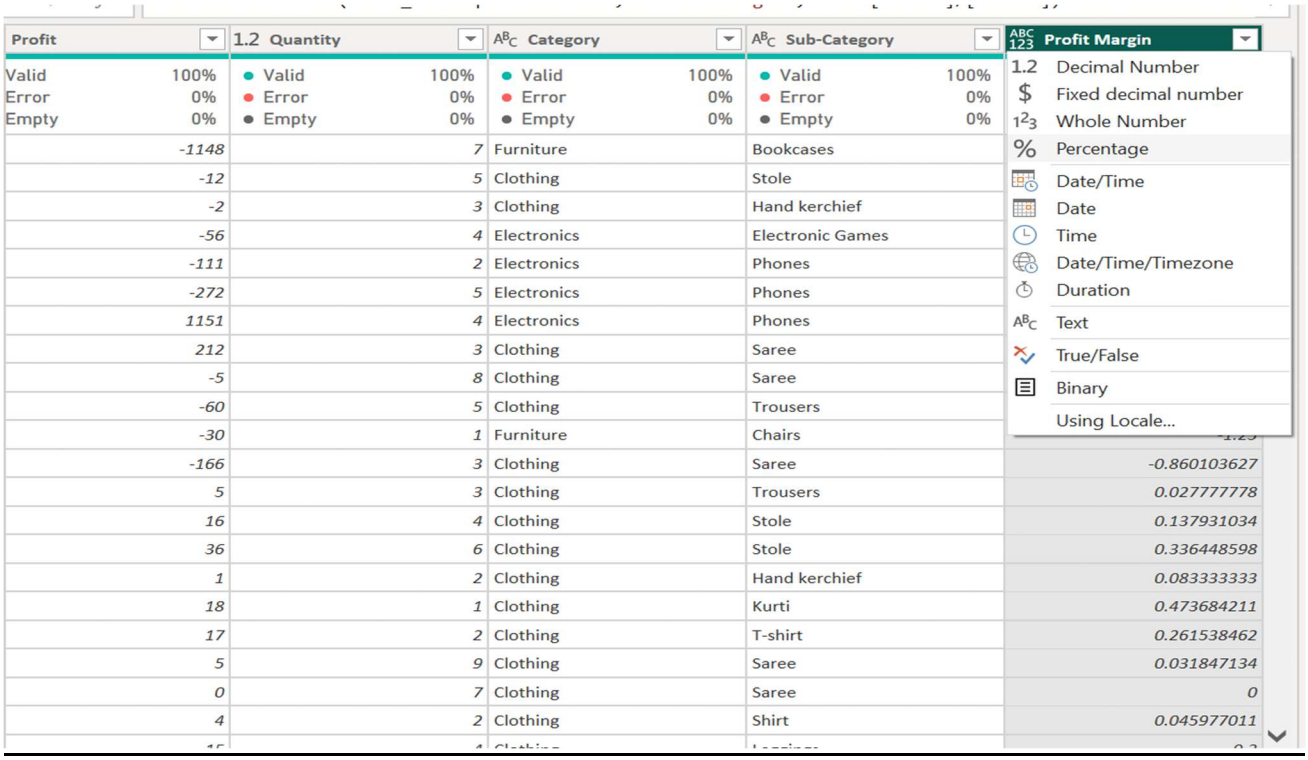


	Order ID	Order Date	CustomerName	State	City
	Valid 100%	Valid 100%	Valid 100%	Valid 100%	Valid 100%
	Error 0%	Error 0%	Error 0%	Error 0%	Error 0%
	Empty 0%	Empty 0%	Empty 0%	Empty 0%	Empty 0%
1	5601	01-Apr-18	Bharat	Gujarat	Ahmedabad
2	5602				
3	5603				
4	5604				
5	5605				
6	5606				
7	5607				
8	5608				
9	5609				
10	5610				
11	5611				
12	5612				
13	5613				
14	5614				
15	5615				
16	5616				
17	5617	17-Apr-18	Sagar	Nagaland	Kohima
18	5618	18-Apr-18	Manju	Andhra Pradesh	Hyderabad
19	5619	18-Apr-18	Ramesh	Gujarat	Ahmedabad
20	5620	20-Apr-18	Sarita	Maharashtra	Pune
21	5621	20-Apr-18	Deepak	Madhya Pradesh	Bhopal
22	5622	22-Apr-18	Monisha	Rajasthan	Jaipur

	CustomerName	Location
	Valid 89%	Valid 100%
	Error 0%	Error 0%
	Empty 11%	Empty 0%
	Bharat	Gujarat-Ahmedabad
	Pearl	Maharashtra-Pune
	Jahan	Madhya Pradesh-Bhopal
	Divsha	Rajasthan-Jaipur
	Kasheen	West Bengal-Kolkata
	Hazel	Karnataka-Bangalore
	Sonakshi	Jammu and Kashmir-Kashmir
	Aarushi	Tamil Nadu-Chennai
	Jitesh	Uttar Pradesh-Lucknow
	Yogesh	Bihar-Patna
	Anita	Kerala -Thiruvananthapuram
	Shrichand	Punjab-Chandigarh
	Mukesh	Haryana-Chandigarh
	Vandana	Himachal Pradesh-Simla
	Bhavna	Sikkim-Gangtok
	Kanak	Goa-Goa
	Sagar	Nagaland-Kohima
	Manju	Andhra Pradesh-Hyderabad
	Ramesh	Gujarat-Ahmedabad
	Sarita	Maharashtra-Pune
	Deepak	Madhya Pradesh-Bhopal
	Monisha	Rajasthan-Jaipur
	Atharv	West Bengal-Kolkata

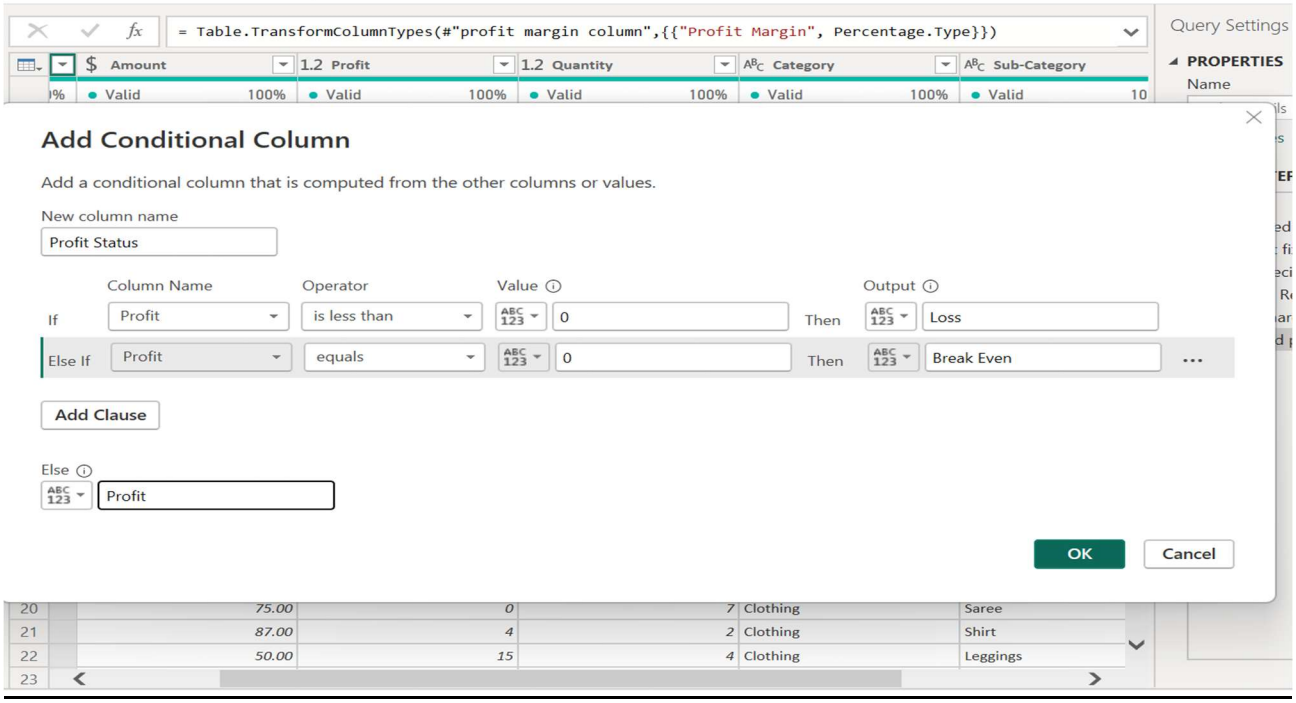
Create a custom column named Profit Margin calculated as: Profit / Amount (expressed as a percentage)



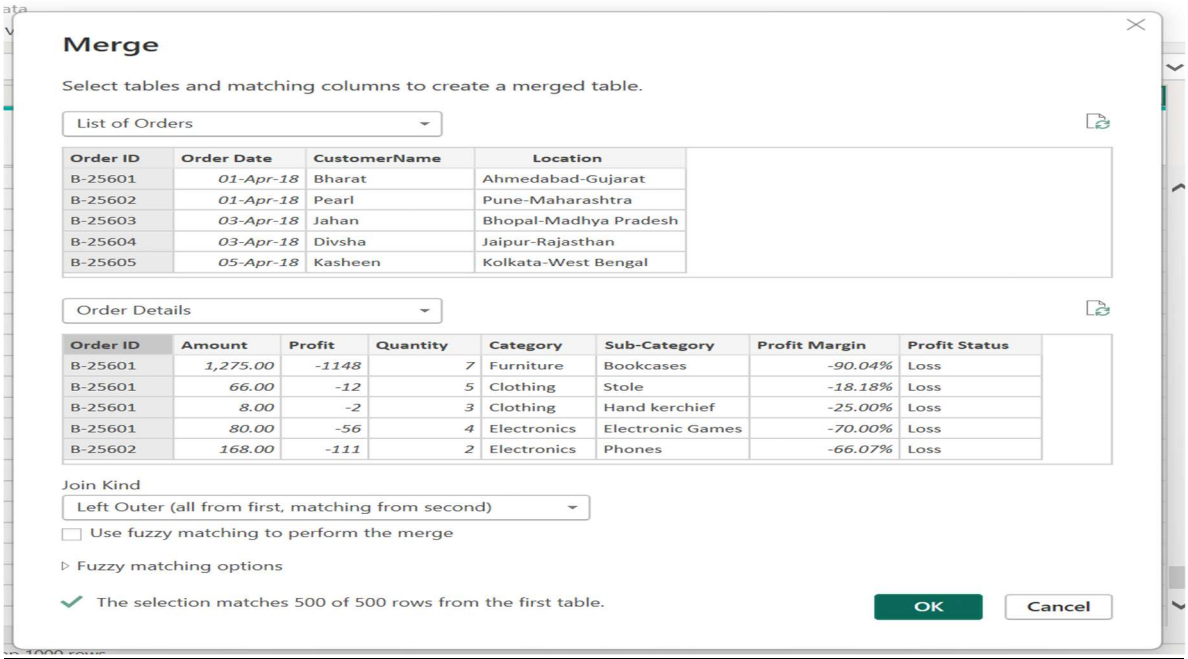


Add a conditional column named Profit Status based on:

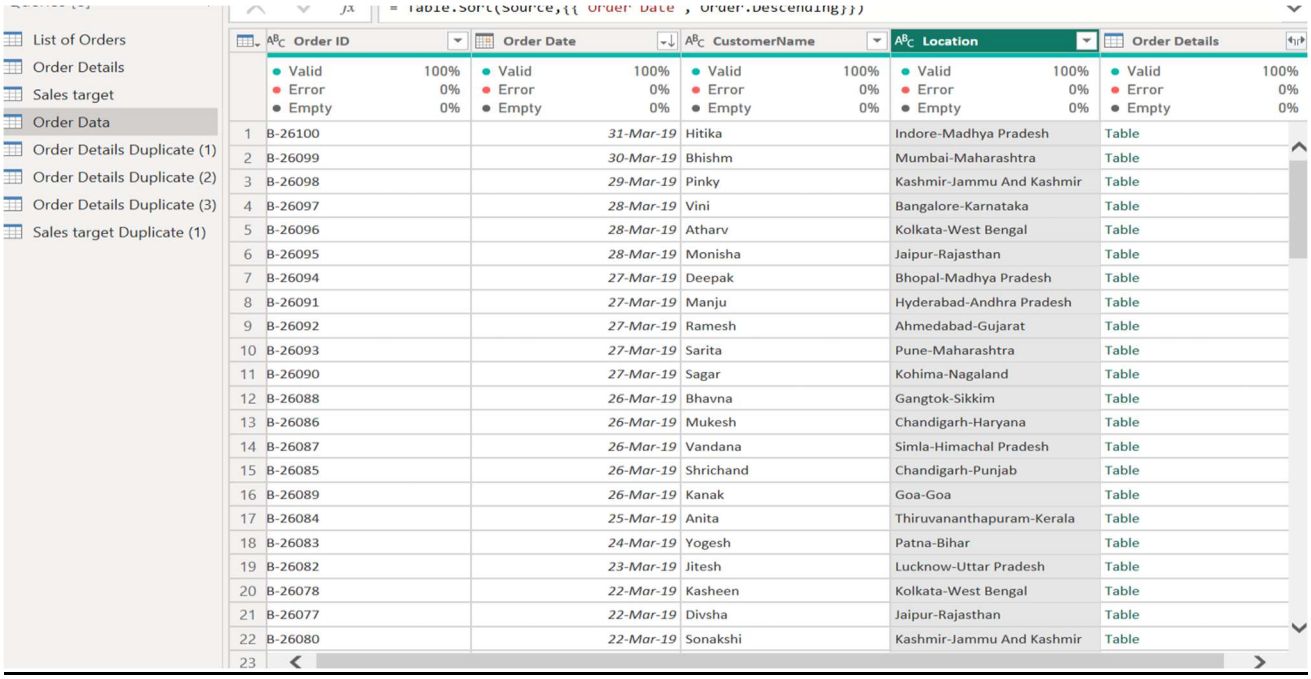
- Profit < 0 → Loss
- Profit = 0 → Break-Even
- Profit > 0 → Profit



Merge the List of Orders and Order Details tables using Order ID:



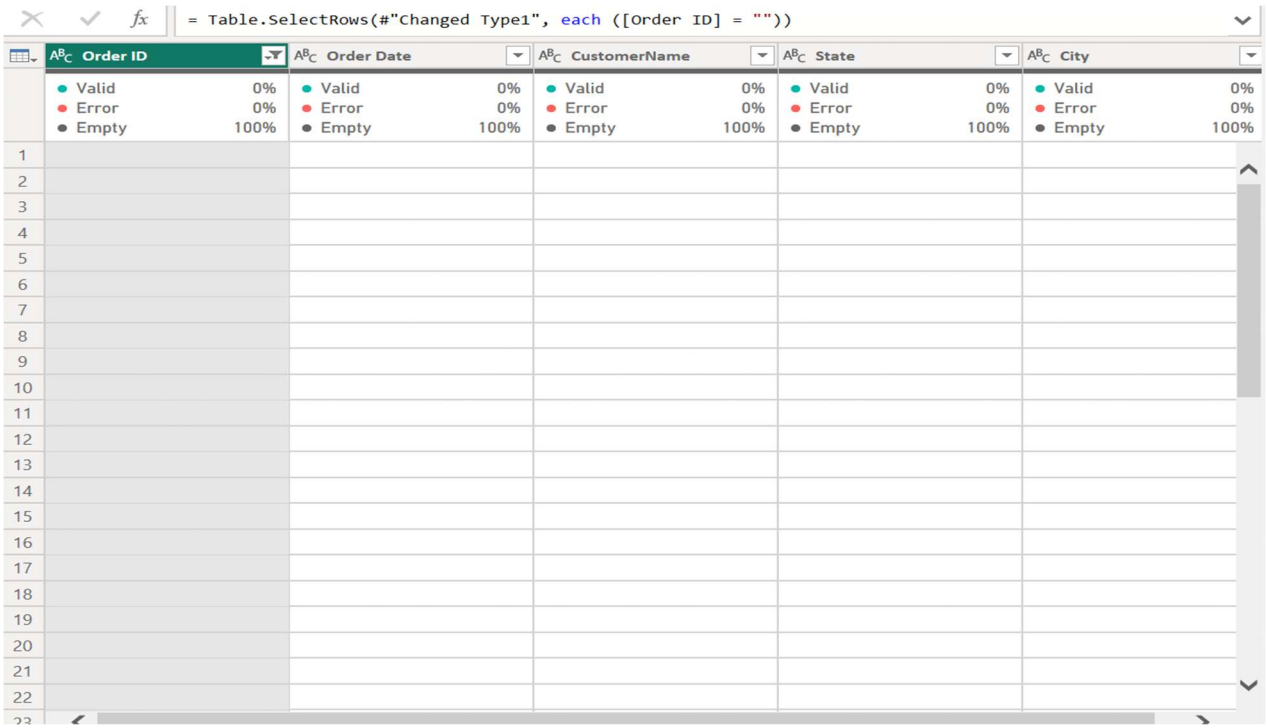
Name the merged output table as Orders Data:



- List of Orders
- Order Details
- Sales target
- Order Data
- Order Details Duplicate (1)
- Order Details Duplicate (2)
- Order Details Duplicate (3)
- Sales target Duplicate (1)

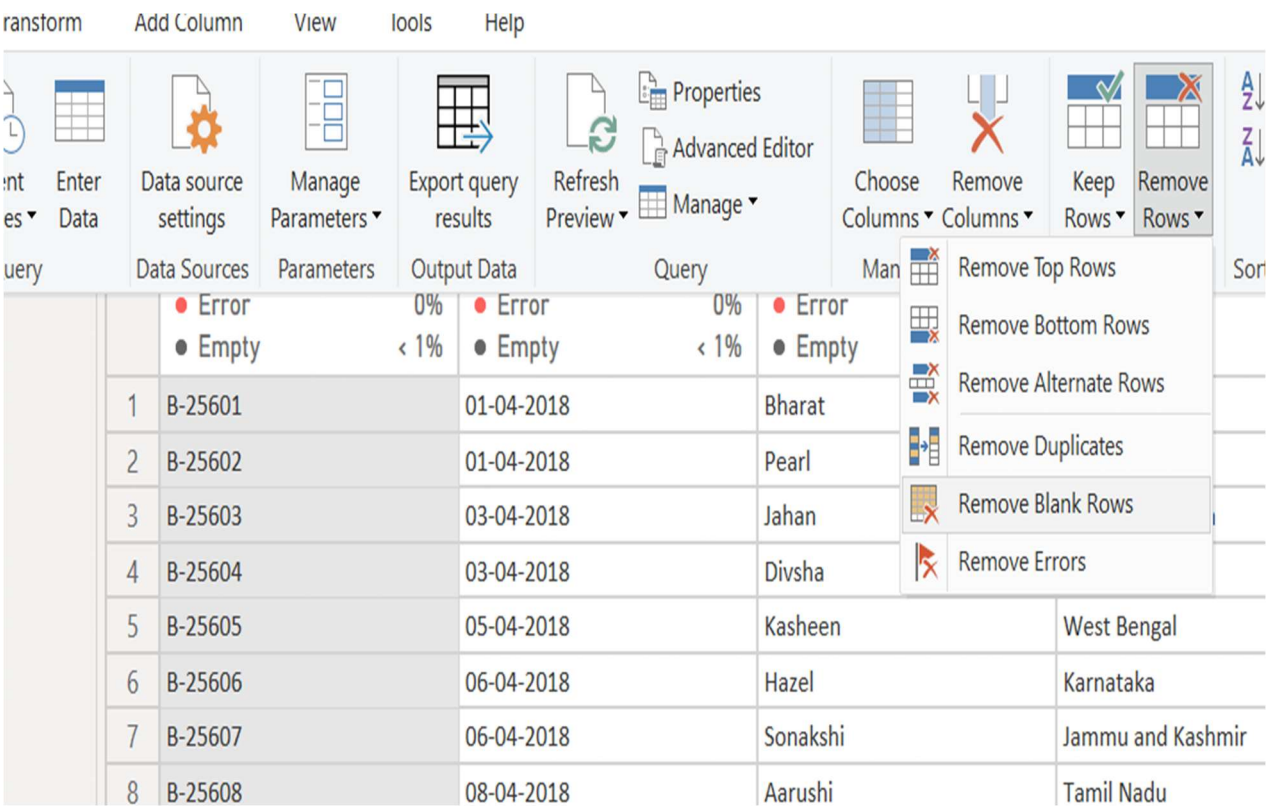
	Order ID	Order Date	CustomerName	Location	Order Details
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-26100	31-Mar-19	Hitika	Indore-Madhya Pradesh	Table
2	B-26099	30-Mar-19	Bhishm	Mumbai-Maharashtra	Table
3	B-26098	29-Mar-19	Pinky	Kashmir-Jammu And Kashmir	Table
4	B-26097	28-Mar-19	Vini	Bangalore-Karnataka	Table
5	B-26096	28-Mar-19	Atharv	Kolkata-West Bengal	Table
6	B-26095	28-Mar-19	Monisha	Jaipur-Rajasthan	Table
7	B-26094	27-Mar-19	Deepak	Bhopal-Madhya Pradesh	Table
8	B-26091	27-Mar-19	Manju	Hyderabad-Andhra Pradesh	Table
9	B-26092	27-Mar-19	Ramesh	Ahmedabad-Gujarat	Table
10	B-26093	27-Mar-19	Sarita	Pune-Maharashtra	Table
11	B-26090	27-Mar-19	Sagar	Kohima-Nagaland	Table
12	B-26088	26-Mar-19	Bhavna	Gangtok-Sikkim	Table
13	B-26086	26-Mar-19	Mukesh	Chandigarh-Haryana	Table
14	B-26087	26-Mar-19	Vandana	Simla-Himachal Pradesh	Table
15	B-26085	26-Mar-19	Shrichand	Chandigarh-Punjab	Table
16	B-26089	26-Mar-19	Kanak	Goa-Goa	Table
17	B-26084	25-Mar-19	Anita	Thiruvananthapuram-Kerala	Table
18	B-26083	24-Mar-19	Yogesh	Patna-Bihar	Table
19	B-26082	23-Mar-19	Jitesh	Lucknow-Uttar Pradesh	Table
20	B-26078	22-Mar-19	Kasheen	Kolkata-West Bengal	Table
21	B-26077	22-Mar-19	Divsha	Jaipur-Rajasthan	Table
22	B-26080	22-Mar-19	Sonakshi	Kashmir-Jammu And Kashmir	Table
23					

Removing records: (Removing Blank rows)

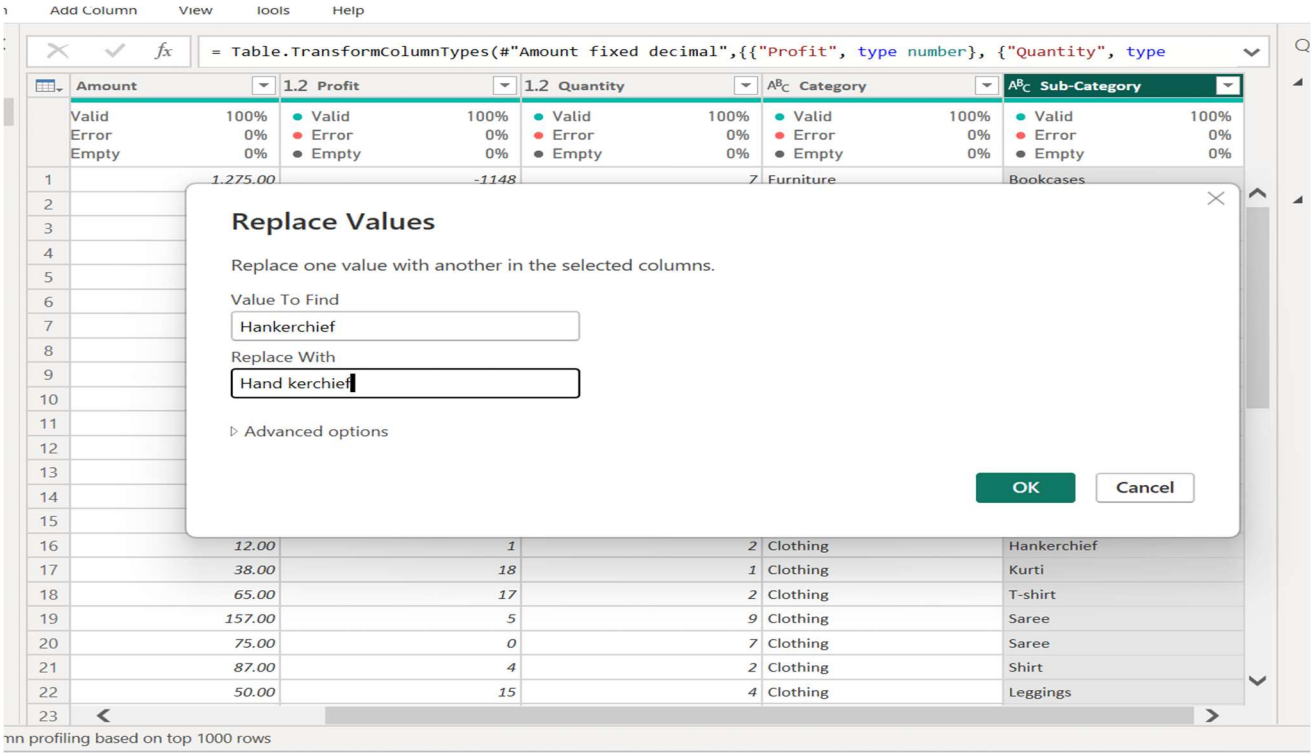


```
= Table.SelectRows(#"Changed Type1", each ([Order ID] = ""))
```

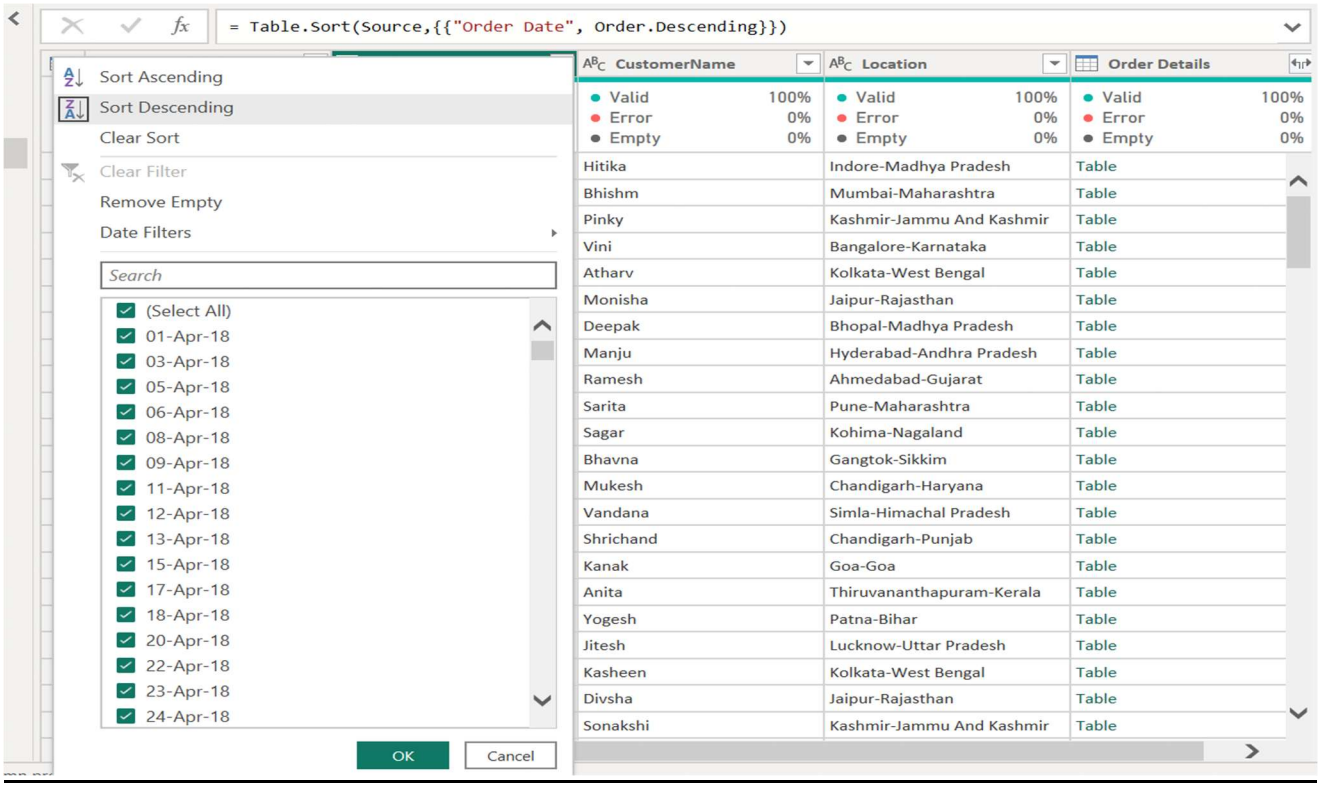
	Order ID	Order Date	CustomerName	State	City
	Valid 0% Error 0% Empty 100%	Valid 0% Error 0% Empty 100%	Valid 0% Error 0% Empty 100%	Valid 0% Error 0% Empty 100%	Valid 0% Error 0% Empty 100%
1					
2					
3					
4					
5					
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22					



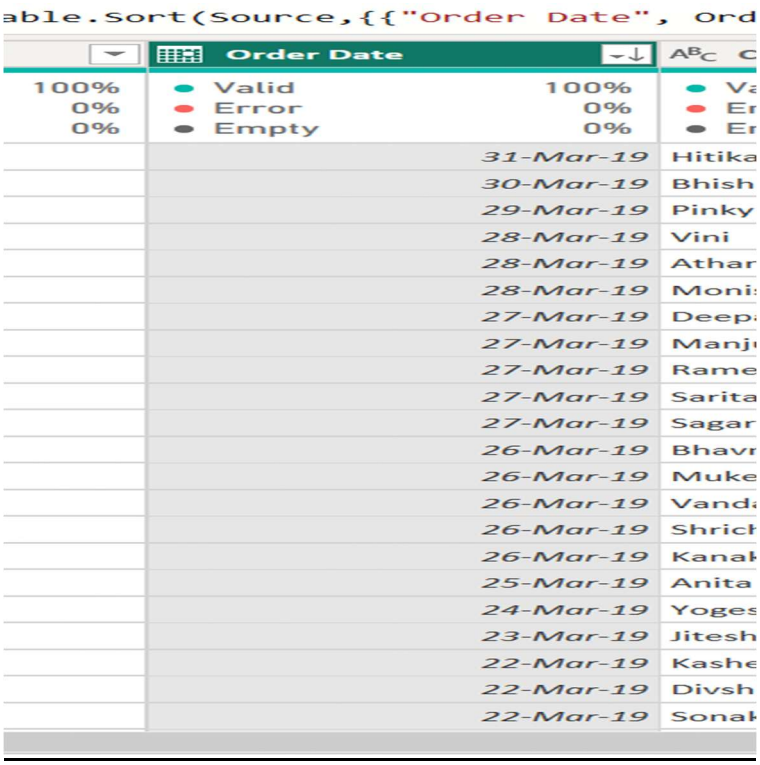
Replace Value in sub_category:



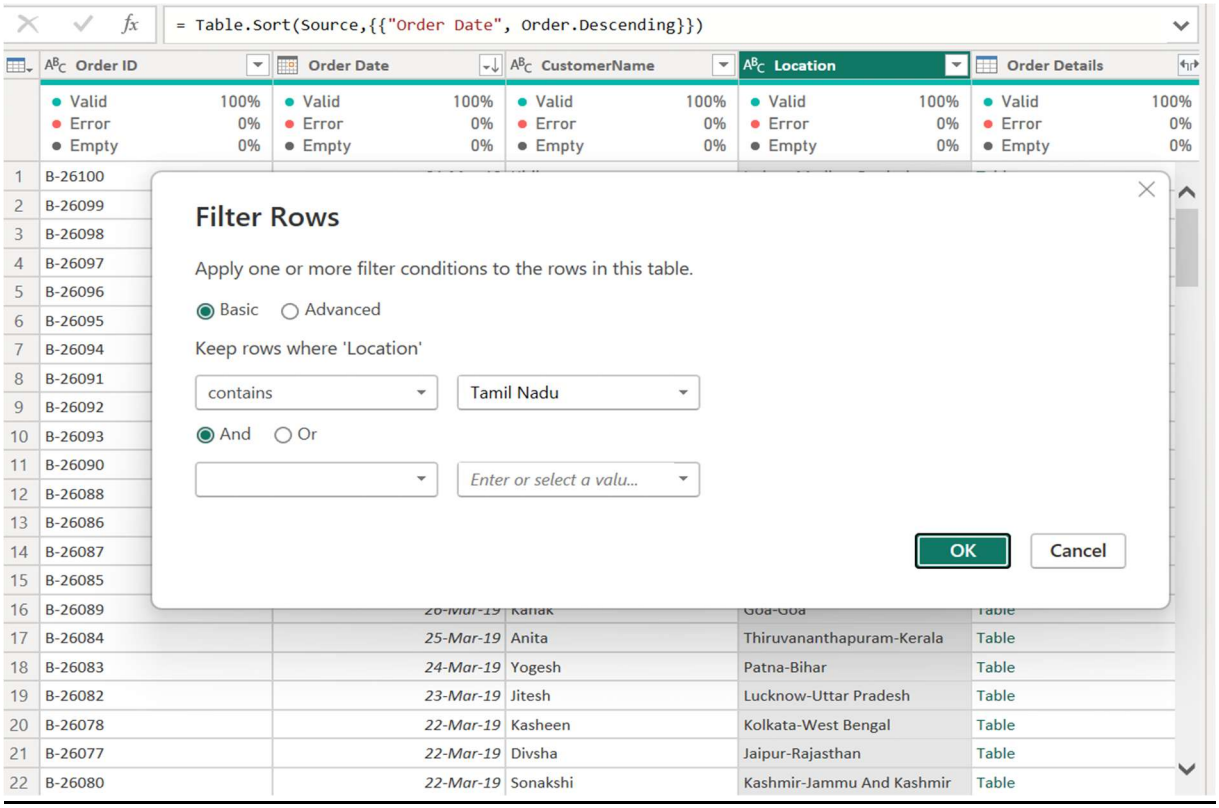
Sort records in the Orders Data table by Order Date in descending order:



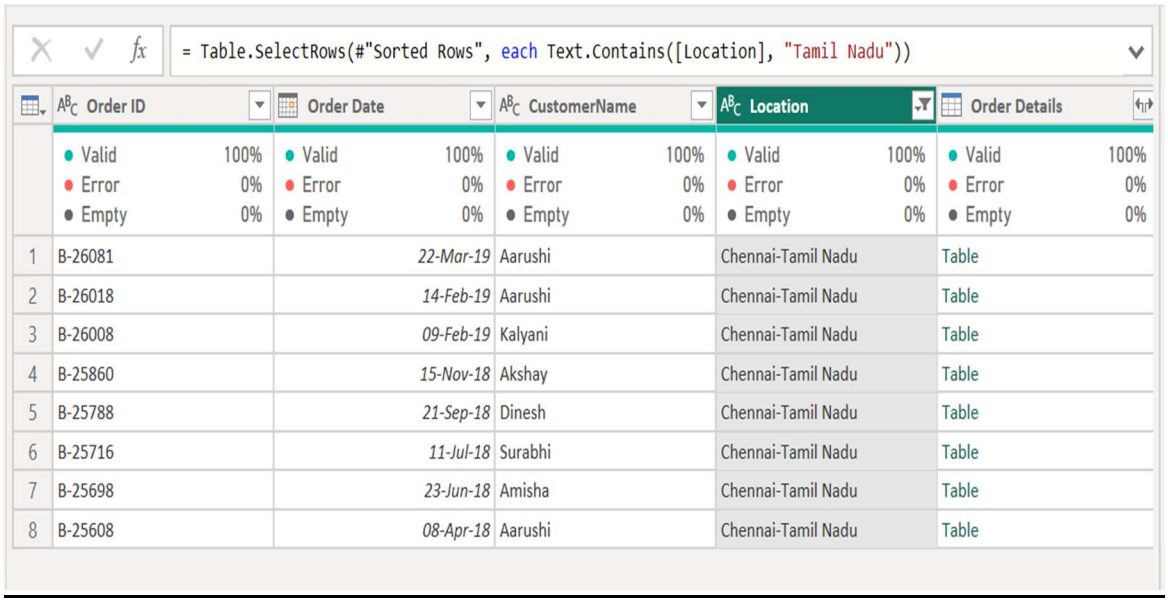
Descending order output:



Apply filters to analyze orders from a specific state Tamil Nadu:



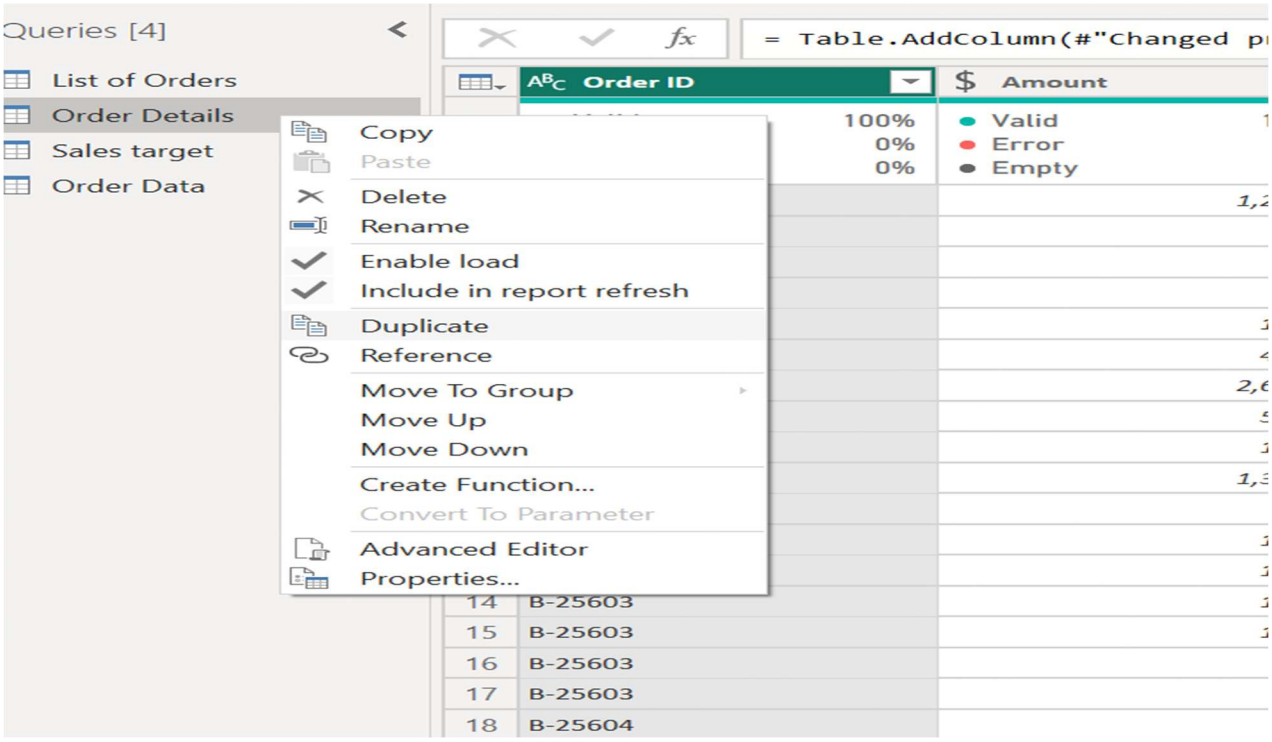
	Order ID	Order Date	CustomerName	Location	Order Details
	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%
1	B-26100				
2	B-26099				
3	B-26098				
4	B-26097				
5	B-26096				
6	B-26095				
7	B-26094				
8	B-26091				
9	B-26092				
10	B-26093				
11	B-26090				
12	B-26088				
13	B-26086				
14	B-26087				
15	B-26085				
16	B-26089	26-Mar-19	Rahak	Goa-Goa	Table
17	B-26084	25-Mar-19	Anita	Thiruvananthapuram-Kerala	Table
18	B-26083	24-Mar-19	Yogesh	Patna-Bihar	Table
19	B-26082	23-Mar-19	Jitesh	Lucknow-Uttar Pradesh	Table
20	B-26078	22-Mar-19	Kasheen	Kolkata-West Bengal	Table
21	B-26077	22-Mar-19	Divsha	Jaipur-Rajasthan	Table
22	B-26080	22-Mar-19	Sonakshi	Kashmir-Jammu And Kashmir	Table



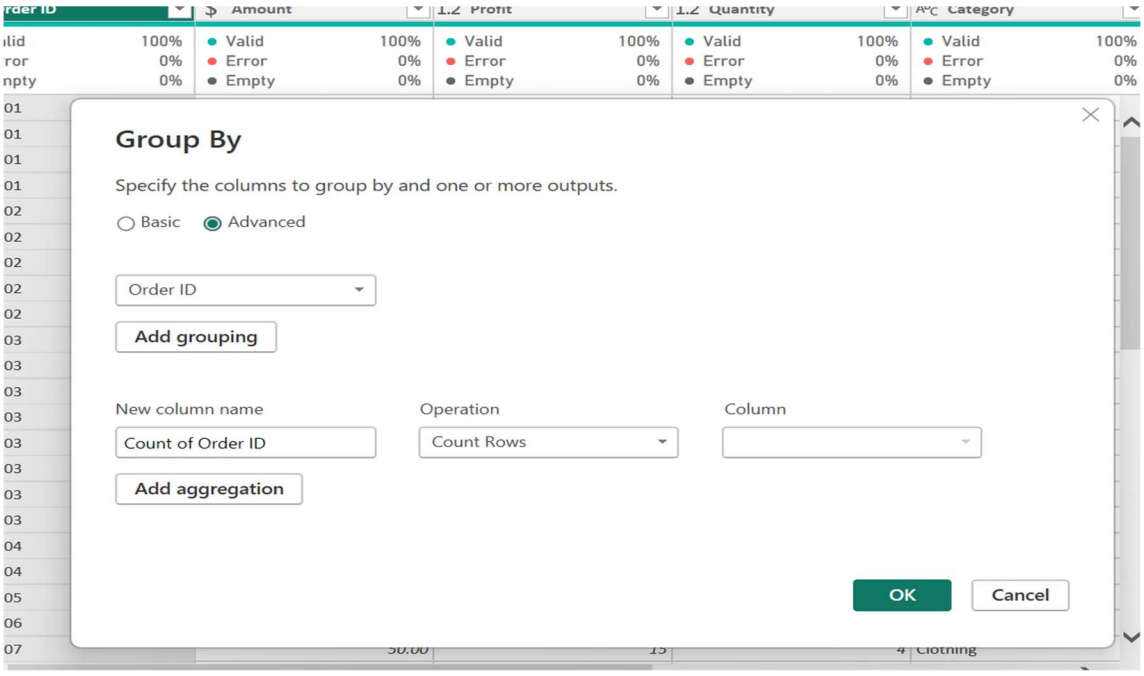
	Order ID	Order Date	CustomerName	Location	Order Details
	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%	Valid 100% / Error 0% / Empty 0%
1	B-26081	22-Mar-19	Aarushi	Chennai-Tamil Nadu	Table
2	B-26018	14-Feb-19	Aarushi	Chennai-Tamil Nadu	Table
3	B-26008	09-Feb-19	Kalyani	Chennai-Tamil Nadu	Table
4	B-25860	15-Nov-18	Akshay	Chennai-Tamil Nadu	Table
5	B-25788	21-Sep-18	Dinesh	Chennai-Tamil Nadu	Table
6	B-25716	11-Jul-18	Surabhi	Chennai-Tamil Nadu	Table
7	B-25698	23-Jun-18	Amisha	Chennai-Tamil Nadu	Table
8	B-25608	08-Apr-18	Aarushi	Chennai-Tamil Nadu	Table

Grouping and Aggregating Data

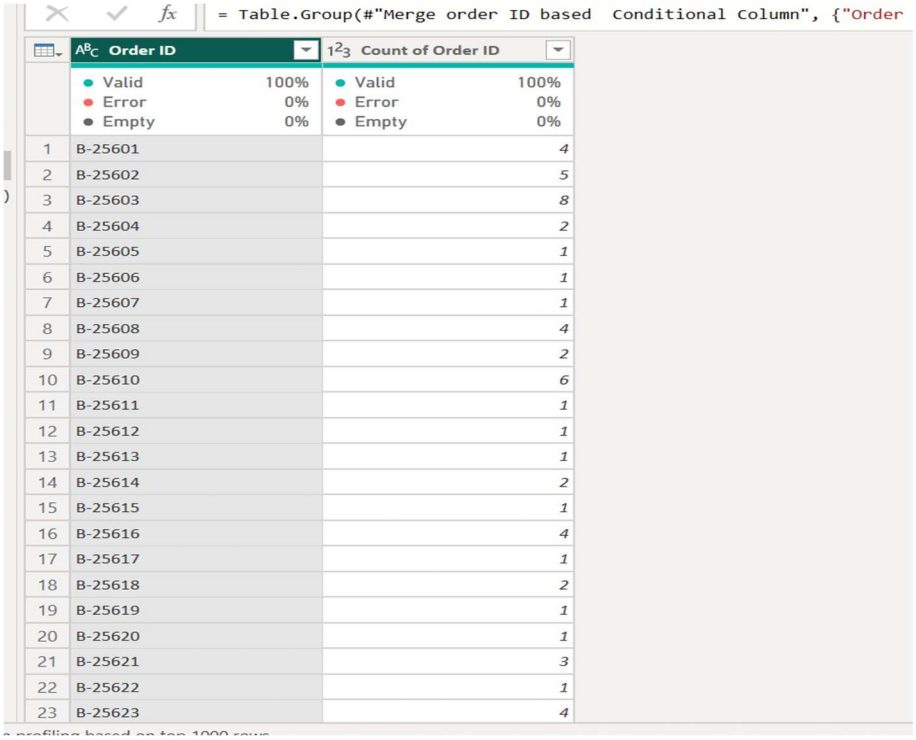
Order Details Table Duplicate:



Count of Order ID:



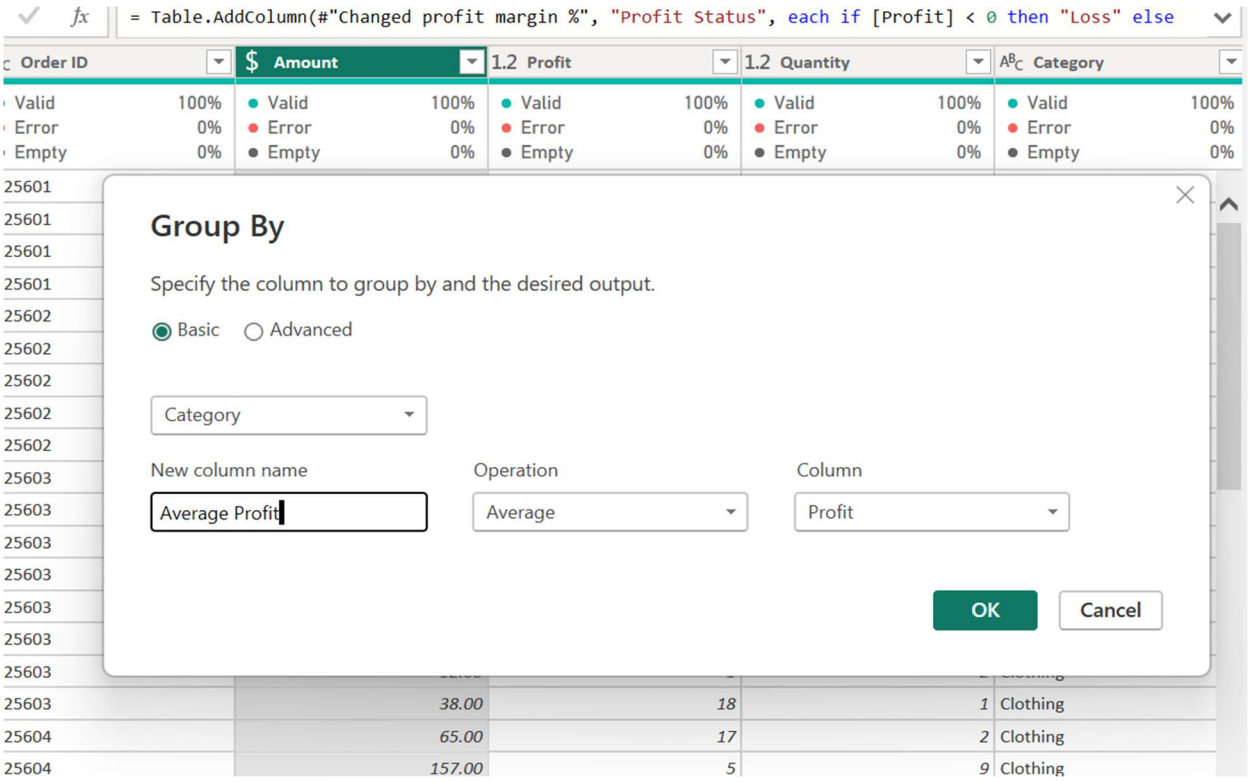
Count of Order ID column create:



```
= Table.Group(#"Merge order ID based  Conditional Column", {"Order
```

	Order ID	Count of Order ID
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	B-25601	4
2	B-25602	5
3	B-25603	8
4	B-25604	2
5	B-25605	1
6	B-25606	1
7	B-25607	1
8	B-25608	4
9	B-25609	2
10	B-25610	6
11	B-25611	1
12	B-25612	1
13	B-25613	1
14	B-25614	2
15	B-25615	1
16	B-25616	4
17	B-25617	1
18	B-25618	2
19	B-25619	1
20	B-25620	1
21	B-25621	3
22	B-25622	1
23	B-25623	4

Average Profit by Category:



```
= Table.AddColumn(#"Changed profit margin %", "Profit Status", each if [Profit] < 0 then "Loss" else
```

Order ID	Amount	Profit	Quantity	Category
Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
25601				
25601				
25601				
25601				
25602				
25602				
25602				
25602				
25602				
25603				
25603				
25603				
25603				
25603				
25603				
25603				
25603	38.00	18	1	Clothing
25604	65.00	17	2	Clothing
25604	157.00	5	9	Clothing

Group By

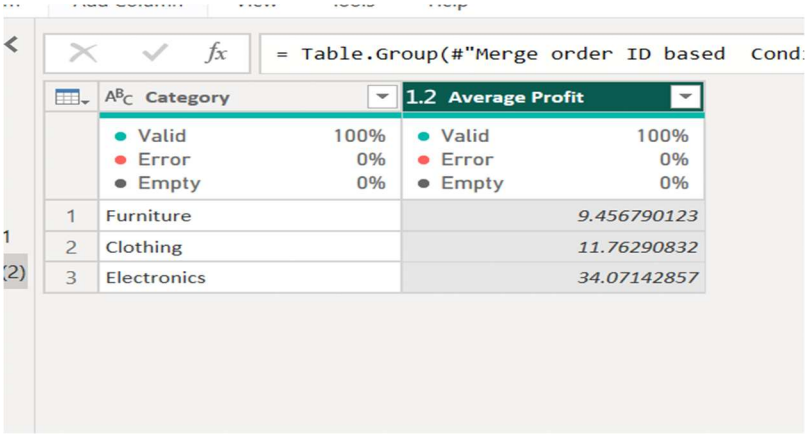
Specify the column to group by and the desired output.

(•) Basic ( ) Advanced

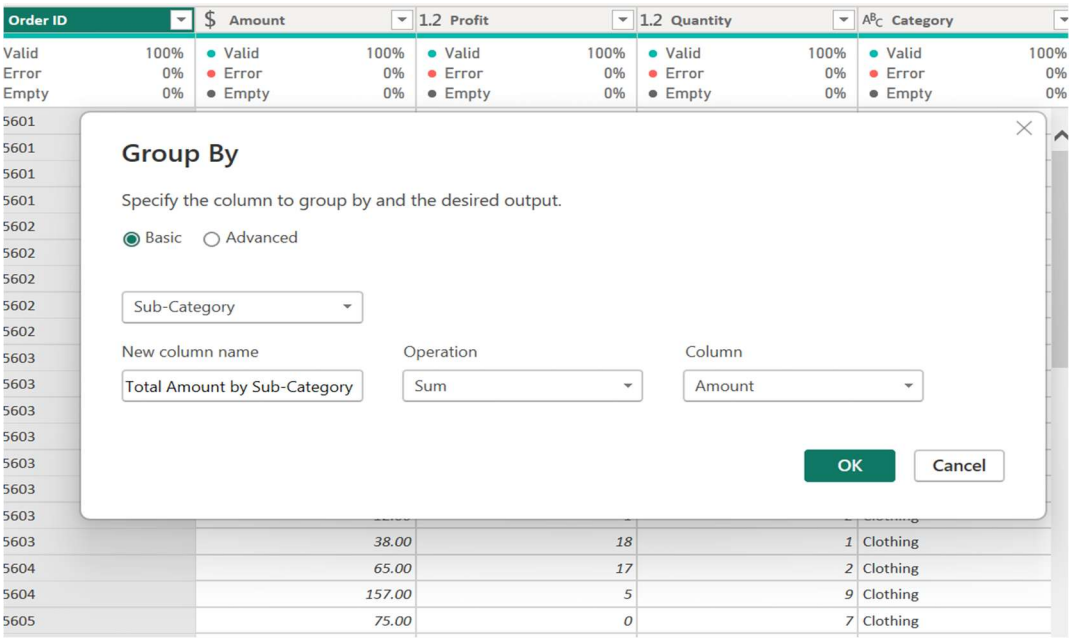
Category

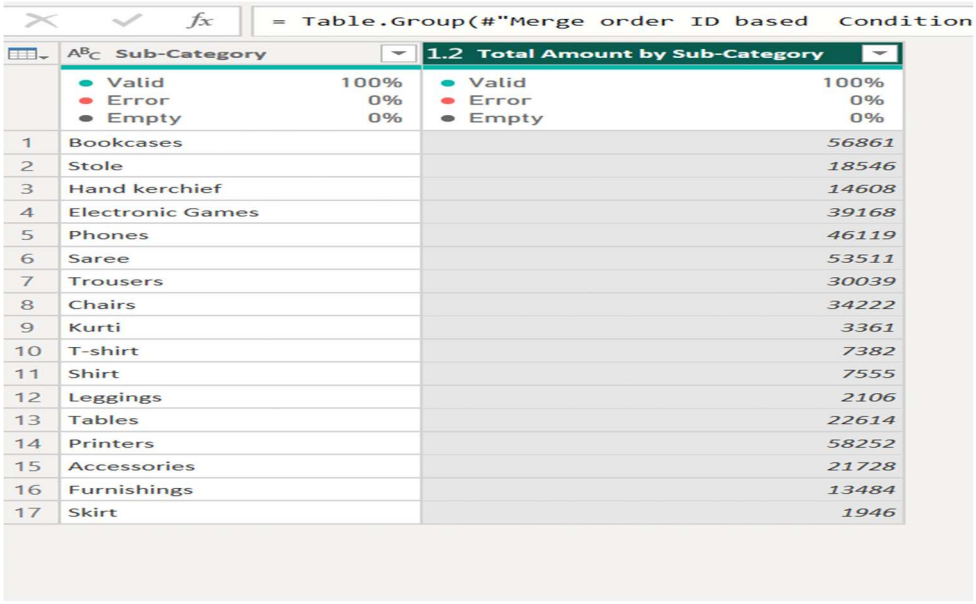
New column name	Operation	Column
Average Profit	Average	Profit

OK Cancel

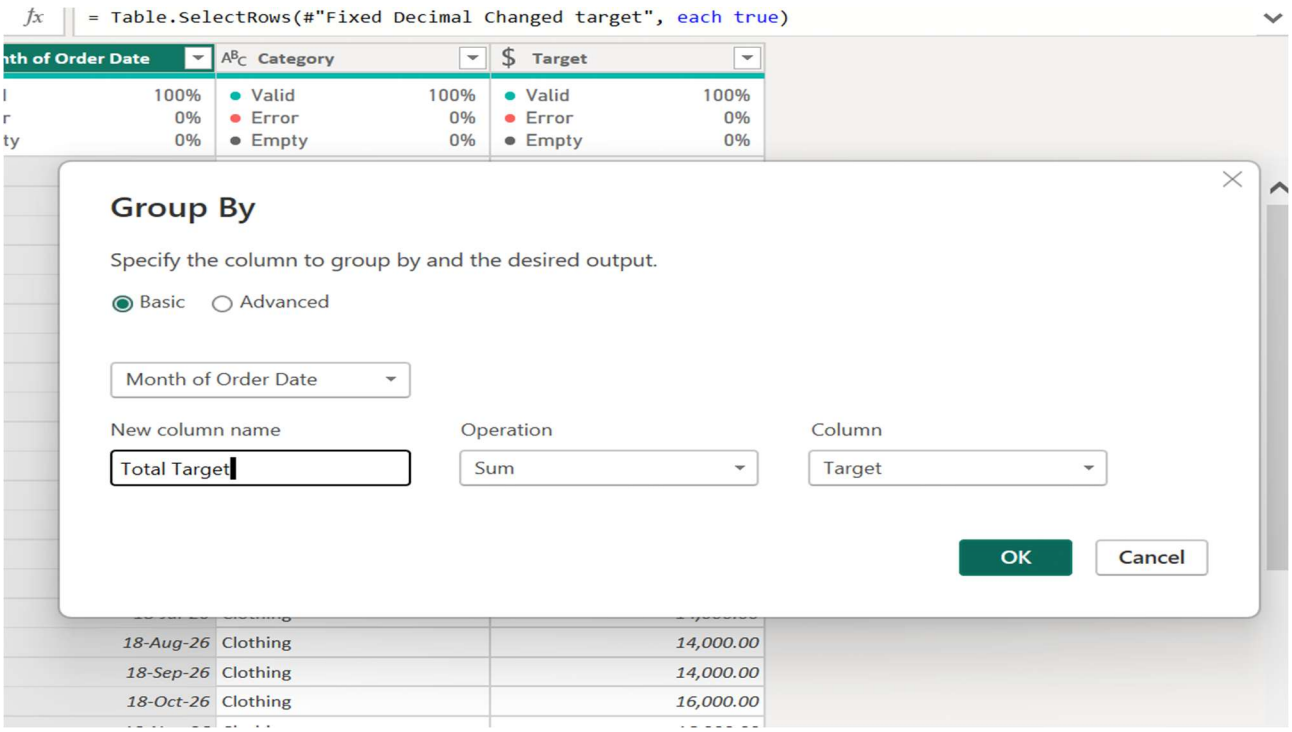


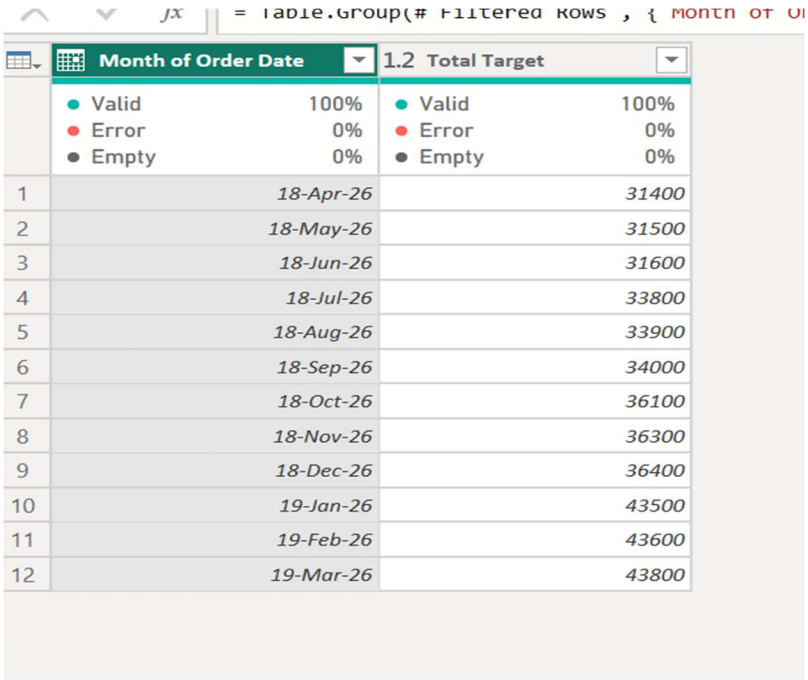
Total Amount by Sub-Category:





Duplicate the Sales Target table and aggregate the total Target by Month of order date:





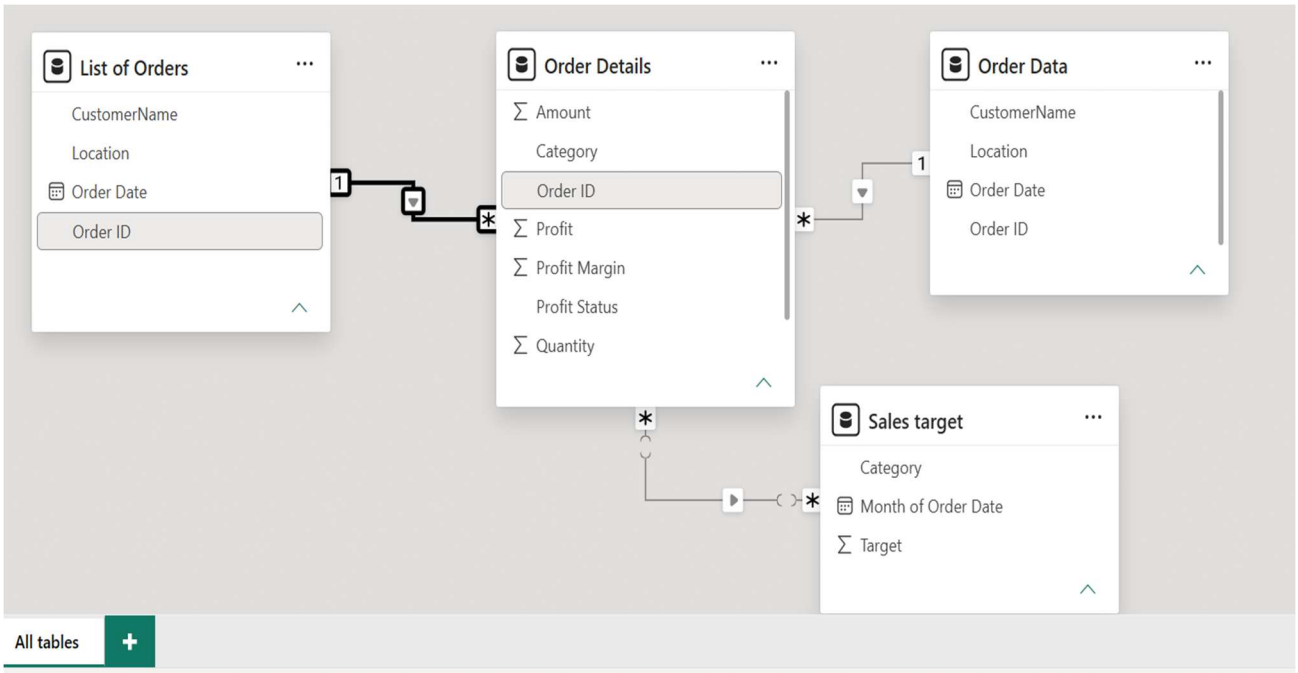
	Month of Order Date	1.2 Total Target
	Valid 100% Error 0% Empty 0%	Valid 100% Error 0% Empty 0%
1	18-Apr-26	31400
2	18-May-26	31500
3	18-Jun-26	31600
4	18-Jul-26	33800
5	18-Aug-26	33900
6	18-Sep-26	34000
7	18-Oct-26	36100
8	18-Nov-26	36300
9	18-Dec-26	36400
10	19-Jan-26	43500
11	19-Feb-26	43600
12	19-Mar-26	43800

Data Modeling

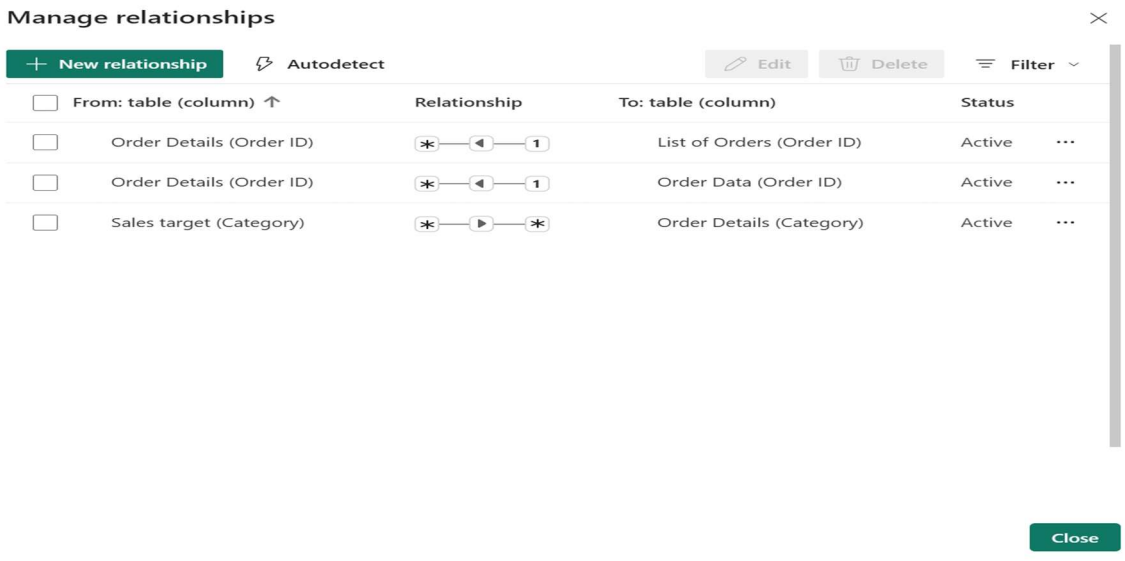
Relationship between List of Orders and Order Details using Order ID

Relationship between Order Details and Sales Target using Category

Relationship between Order Details and Order Data using Order ID



Manage Relationships to ensure relationships are active and correctly defined



Brief explanation of transformation and modeling logic

1. Data Import & Transformation Logic (Power Query)

The raw CSV files (List of Orders, Order Details, and Sales Target) underwent extensive cleaning, formatting, and feature engineering to ensure data integrity before reporting:

- **Data Type Definitions:** Standardized column types, such as setting Order Date to a proper Date type using locale formatting, and converting financial fields (Amount, Target) into fixed decimal/currency formats.
- **Data Cleaning & Text Formatting:** Standardized text strings by trimming whitespace, correcting typos (e.g., changing "Hankerchief" to "Hand kerchief"), and formatting customer names into proper case.
- **Calculated & Conditional Columns:**
    - **Location:** Merged the City and State columns into a single descriptive location attribute.
    - **Profit Margin:** Created a custom column dividing Profit by Amount and formatting it as a percentage.
    - **Profit Status:** Built a conditional column categorizing orders into Loss, Break-Even, or Profit based on profit thresholds.
- **Table Merging & Aggregation:** Combined related tables using Order ID (e.g., creating the Orders Data table) and built summarized views using Group By operations (e.g., Average Profit by Category, Total Amount by Sub-Category, and Total Target by Month).

2. Data Modelling Logic:

Once the transformed tables were loaded into the data model, relationships were established to connect transactional details with dimensional data:

- **List of Orders to Order Details:** Linked via Order ID in a one-to-many relationship, connecting order-level metadata (customer, location, date) to line-item specifics.
- **Order Details to Order Data:** Connected via Order ID to support consolidated analysis.
- **Sales Target to Order Details:** Linked via the Category field to map category-level targets against actual sales performance.
- **Relationship Management:** Verified and maintained active relationships in the model to ensure accurate cross-filtering and measure calculations across visuals.